Karthik Mahadev B

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Summary

Full Stack Developer with 2+ years of experience building and deploying production web and mobile applications using React.js, Next.js, TypeScript, Node.js, Express.js, and PostgreSQL. Experienced in end-to-end product development, REST API integrations, authentication, payment workflows, database design, performance optimization, and cloud deployment. Built production e-commerce and multi-tenant SaaS platforms, along with an AI-powered chat application integrating Google Gemini, tool calling, and external APIs. Hands-on experience using AI coding tools to support development, debugging, and problem-solving, with a focus on delivering reliable, business-oriented applications.

Experience

Freelance – **Full Stack Developer** **August 2026 – Present**

The Relish Jar – Production E-Commerce Platform

Kridhan's Enterprises, Coimbatore

- Independently designed, developed, and deployed a production e-commerce platform using Next.js, TypeScript, Tailwind CSS, and Supabase/PostgreSQL, owning requirements, frontend, backend workflows, integrations, and deployment.
- Designed PostgreSQL data models and Row-Level Security (RLS) policies for products, carts, orders, and customer records, enforcing separation between public and protected data.
- Integrated Razorpay with server-side order creation, webhook signature verification, and idempotent order processing for reliable production payments.
- Developed an authenticated admin dashboard for product, order, and fulfillment management, enabling the client to operate the store independently.
- Integrated Google Analytics for comprehensive user behavior tracking and e-commerce performance monitoring.
- Implemented transactional email notifications and GitHub-based CI/CD, deploying production releases through Vercel.

BinaryBudds – **Founding Full Stack Developer** **Dec 2025 – July 2026**

BB FleetTracker – Live GPS Fleet-Tracking Platform

- Built and shipped a multi-tenant GPS fleet-tracking SaaS from the ground up using Next.js, TypeScript, Supabase/PostgreSQL, serving drivers, parents, and school administrators across multiple schools.
- Designed JWT-based authentication, role-based access control, and school-level Row-Level Security (RLS) for tenant data isolation.
- Diagnosed and resolved a production authentication issue involving cookie vs. Bearer-token precedence that was preventing mobile users from authenticating correctly.
- Developed core React Native/Expo features for driver and parent applications and shipped the application to the Google Play Store.
- Implemented production observability using Sentry and Grafana Cloud, and integrated Playwright into CI for automated testing.
- Diagnosed and fixed a production memory leak on DigitalOcean infrastructure, restoring stable long-running application performance.

Origa Technologies – **Software Engineer** **June 2024 – September 2025**

Origa.market – Production E-Commerce Platform

- Developed and optimized a production-scale e-commerce frontend using React.js, Tailwind CSS, Bootstrap,

GraphQL, and Algolia across the product catalog and transactional workflows.

- Built cart, checkout, and PhonePe payment workflows supporting guest and authenticated users with persistent sessions.
- Improved initial page load performance by 35–40% through lazy loading, code splitting, and asset optimization, improving Core Web Vitals.
- Established a responsive, mobile-first UI architecture across product, cart, checkout, and transactional workflows.
- Improved frontend reliability by expanding Jest test coverage, eliminating render-blocking resources, and remediating security vulnerabilities.

Origa Financial Intelligence Dashboard

- Built a Next.js financial intelligence dashboard for machine valuation, loan tenure, and interest-rate scenario analysis.
- Automated financial calculations, reducing manual evaluation effort by 50%.

Projects

FinSight – Full Stack Financial Analytics Platform *Live Project*

- Built a full-stack expense analytics platform using Next.js, Node.js, and PostgreSQL, categorizing transactions, analyzing spending patterns, and generating actionable financial insights.
- Implemented JWT-based authentication and per-user data isolation to protect financial records.
- Deployed frontend and backend to production with secure environment configuration and a performance-tuned, responsive UI.

Symmetra AI — AI-Powered Chat Application *Live Project*

- Built and deployed an AI-powered chat application using Next.js, TypeScript, Google Gemini API, and PostgreSQL.
- Implemented LLM tool-calling workflows that connect AI responses to external APIs, including weather, Formula 1, and stock-related functionality.
- Developed persistent conversation history, streaming responses, OAuth authentication, and protected application routes.
- Integrated database operations and external service calls into the application, connecting the chat interface with backend functionality.

Education

B.E Mechanical Engineering **Aug 2019 – June 2023**
Ramco Institute of Technology - Rajapalayam, India

- CGPA: 7.7

Skills

Programming Languages: JavaScript, TypeScript, SQL

Frontend Technologies: React.js, Next.js, React Native, HTML5, CSS3, Tailwind CSS, Redux

Backend Technologies: Node.js, Express.js, REST APIs

Databases: PostgreSQL, Supabase, MongoDB

AI & AI-Assisted Development: Google Gemini API, LLM Tool Calling, AI API Integration, ChatGPT, Claude Code, GitHub Copilot

Cloud & DevOps: Vercel, AWS , Render , DigitalOcean, Docker, GitHub, CI/CD

Testing & Observability: Playwright, Sentry, Grafana Cloud

Integrations & Tools: GraphQL, Razorpay, PhonePe, Algolia, Git, Android Studio